Unwrapping the mind:
knowledge, cognition, and AI

Hello there!

What are your expectations?

UNIVERSITÄT MANNHEIM

QR Code #725330

Let's brainstorm a bit!

What are your associations to *"Unwrapping the mind: knowledge, cognition and AI"*?

Unwrapping the mind: knowledge, cognition, and AI

Lecture topic / overview

What this lecture is about

(From the module description: let's take a closer look at it together, paragraph by paragraph!)

With the rise of AI models like GPT-4, machines seem capable of mimicking human cognitive processes-thinking, perceiving, learning, and even making decisions. But ***do they truly think, or are they merely processing patterns in complex ways***?

What this lecture is about

In this lecture series, experts in cognitive science will explore **what makes human cognition exceptional (if at all), how AI challenges our understanding of intelligence, and what we can gain from integrating AI into cognitive research**. We will also address how insights from human cognition can help develop more efficient, interpretable AI models and examine the ethical considerations that come with their use.

What this lecture is about - Psychology

The perspective of psychology will approach cognition using both experiments and formal modeling. Lectures will tackle **human judgment and decision making, memory processes and memory strategies developing across the life-span, categorization and estimation as well as metacognition** - the knowledge about one's own cognitive processes.

Lecturers - Psychology

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Arndt Bröder

- Chair of Experimental Psychology since 2010
- Habilitation 2005 in Bonn
- Ph.D. 1999 in Heidelberg
- Diploma Psychology 1995 Bonn

Research Interests:

- Judgment and Decision Making
- (Source) Memory and Metamemory
- Methodology and Modeling

Lecturers - Psychology

Beatrice G. Kuhlmann

- Chair of Cognitive Psychology and Cognitive Aging
- Mannheim Cognitive Aging Lab

Research Interests

- Episodic long-term memory, especially source memory
- Cognitive aging
- Model-based separation of memory from guessing
- Metacognition (judging your memory, encoding strategies)

Background

- Cognitive & Developmental Psychology

PhD graduation, 2013
University of North Carolina at Greensboro

About 10 years later...

What this lecture is about - Linguistics

From the linguistic perspective, we will **focus on language acquisition and processing, as well as on the question how language relates to other domains of cognition**. We will present both experimental results and historical corpus data to showcase how language is represented and processed in the mind and how learners acquire language along the dimension of time.

Lecturers - Linguistics

Nicole Altvater-Mackensen

- Chair of Psycholinguistics
- Child Research Lab *Wortakrobaten*

Research Interests

- Language acquisition, representation & processing
- Interplay of language and other domains of cognition

Background

- Linguistics, Developmental & Cognitive Psychology, Neuroscience

Lecturers - Linguistics

Carola Trips

- Chair of English Linguistics/Diachrony
- Host of the [Penn Historical Corpora] website

Research Interests

- Language change, language contact, language acquisition,
- linguistic theory, corpus linguistics

Background

- Linguistics of English and German

Lecturers - Linguistics

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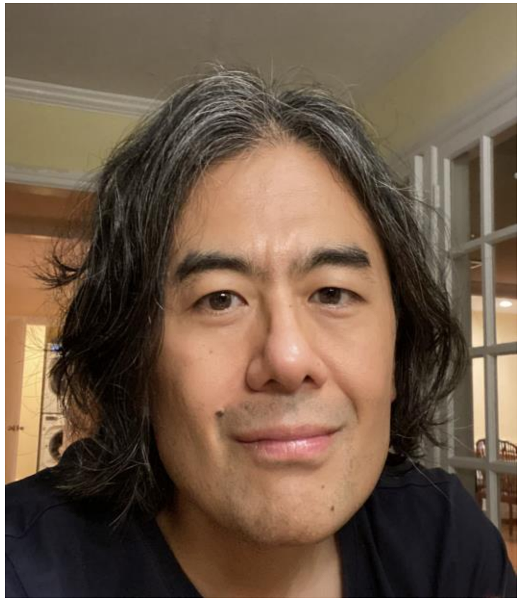
Charles Yang

- Professor of Linguistics, Computer Science and Psychology
- Director, Program in Cognitive Science
- University of Pennsylvania, Philadelphia USA



Research Interests

- Language learning, variation and change
- Conceptual and cognitive development
- Computational social science

Background

- Ph.D. in computer science (MIT 2000)

What this lecture is about – Computational Science

From a computational and AI-driven standpoint, the series will cover the **modeling of cognitive processes, investigating how human judgment and decision-making are represented in computational frameworks**.

We will also explore **the inner workings of AI language models, discussing how they process and generate language and how they can be extended through multimodal learning and interaction with users**.

Lecturer – Computational Science

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Simone Ponzetto

- Chair of Information Systems III
- NLP and IR group @ UniMa
- Data and Web science group

Research interest

- Artificial Intelligence
- Language and computation

… and their application in

– (Digital) Humanities
– (Computational) Social Science

Background: AI, NLP, linguistics

What this lecture is about – Philosophy

The philosophy lectures will explore **what it means to say that something or someone "thinks" or "feels".** We examine whether attributing mental states to artificial machines differs fundamentally from doing so to humans or non-human animals.

Lecturer – Philosophy

Wolfgang Freitag

- Chair of Theoretical Philosophy/Philosophy of Language

Research Interests

- Philosophy of Language/Mind: Pragmatics, Psychopragmatics
- Theoretical Rationality: Norms of Belief
- Confirmation Theory

Bringing it all together…

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We hope that this will also shed light on the **broader implications of AI for our general understanding of mind and agency**. Bringing together research from psychology, linguistics, computational science, and philosophy, this series offers **a rich and interdisciplinary exploration of intelligence, learning, and perception-both human and artificial**.

Unwrapping the mind: knowledge, cognition, and AI

Lecture structure

Lecture Structure

Introduction	
3/9/2025	Unwrapping the mind: lecture introduction
Block 1: Psychology	
10/9/2025	“Knowledge representation and processing models in Cognitive Psychology” (Arndt)
17/9/2025	"The psychology of cognitive aging: Beyond old, slow computers" (Beatrice)
24/9/2025	"Metacognition: A capacity that makes us smart" - Monika Undorf, (TU Darmstadt)
1/10/2025	"A common framework to study the evolution and development of human communication" (Manuel Bohn, Leuphana Lüneburg)

Lecture Structure

Block 2: Linguistics	
8/10/2025	"About statistics and hierarchies in language and beyond" (Nicole)
15/10/2025	Learnability and historical linguistics: Carola
22/10/2025	Learning: Charles
Block 3: Philosophy	
29/10/2025	Philosophy: "Can Humans Think?" (Wolfgang)
5/11/2025	Philosophy: 'AI and the Intentional Stance’ (Thomas Raleigh, Luxembourg)

Lecture Structure

	Block 4: Computational Science
12/11/2025	Computational models: Simone
19/11/2025	Morality in Brains, Minds, and Machines (Frederic R. Hopp, ZPID)
	Grand Finale
26/11/2025	Final discussion (panel discussion)

Unwrapping the mind: knowledge, cognition, and AI

Lecture formalities

Diverse topics – diverse backgrounds...

- This is an interdisciplinary lecture: Not only lecturers but also students come from diverse study backgrounds.
    - Bachelor Culture & Economics (IKW module)
    - Master Psychology (AWG & KPPT)
    - Master of Education
    - Bachelor Current English Literature and Linguistics
    - BA Germanistik, Geschichte, MKW
    - CDSS – psychology track
    - Master Wirtschaftsinformatik and Data Science

… but only one Prüfungsleistung - written exam

- 90 minutes
- Forced choice
- Bring your own device/digital exam
- Scheduled in the exam period
- Three questions on each lecture
- Questions based on lecture content and background text
- You need to correctly answer 50% of questions to pass.

Thank you! Questions?